

Lecture 4 - Continuum Limit

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Abstract

In this lecture we will understand how to make sense of making measurements on a discrete spacetime lattice, and the (renormalisation group) procedure to remove the lattice regularization to make a connection to continuum physics. We will also see why the continuum limit corresponds to a second-order phase transition.

1 Overview

Brief motivation: why do we regularise QFT on a lattice, and what price do we pay?

- The lattice spacing a acts as a UV regulator: momenta are cut off at π/a .
- Physical observables must be independent of a — this is the content of the continuum limit.
- The lattice breaks Lorentz/Poincaré symmetry; we must verify its restoration as $a \rightarrow 0$.

1.1 The lattice action

Define the d -dimensional hypercubic lattice $\Lambda = (a\mathbb{Z})^d$ with sites $x = a\hat{n}$, $\hat{n} \in \mathbb{Z}^d$. For a scalar field $\phi_x \equiv \phi(x)$:

$$S_{\text{lat}}[\phi] = a^d \sum_{x \in \Lambda} \left[\frac{1}{2} \sum_{\mu=1}^d \left(\frac{\phi_{x+\hat{\mu}} - \phi_x}{a} \right)^2 + \frac{m_0^2}{2} \phi_x^2 + \frac{\lambda_0}{4!} \phi_x^4 \right]. \quad (1)$$

Recover the continuum action formally as $a \rightarrow 0$.

1.2 Observables and correlation functions

The partition function and n -point functions:

$$Z = \int \mathcal{D}\phi e^{-S_{\text{lat}}[\phi]}, \quad G^{(n)}(x_1, \dots, x_n) = \frac{1}{Z} \int \mathcal{D}\phi \phi_{x_1} \cdots \phi_{x_n} e^{-S_{\text{lat}}[\phi]}. \quad (2)$$

- Masses extracted from exponential decay of two-point function: $G^{(2)}(x) \xrightarrow{|x| \rightarrow \infty} A e^{-m_{\text{phys}}|x|}$.
- The correlation length $\xi = 1/(m_{\text{phys}}a)$ (in lattice units).

2 Renormalisation Group and the Phase Transition

2.1 Block-spin / Kadanoff blocking

Introduce the RG transformation $\mathcal{R}$: coarse-grain by a factor $b > 1$, integrating out short-distance degrees of freedom.

$$e^{-S'[\phi']} = \int \mathcal{D}\phi T[\phi', \phi] e^{-S[\phi]}, \quad (3)$$

where $T[\phi', \phi]$ is the blocking kernel. The new lattice spacing is $a' = b a$.

2.2 RG flow and fixed points

Write the action in terms of a (possibly infinite) set of couplings $\{g_i\} = (m^2, \lambda, c_1, c_2, \dots)$. Under one RG step:

$$g_i \longrightarrow g'_i = \mathcal{R}_i(\{g_j\}). \quad (4)$$

- **Fixed point** $\{g_i^*\}$: $g_i^* = \mathcal{R}_i(\{g_j^*\})$. At a fixed point the theory is scale-invariant.
- **Relevant directions**: couplings that grow under RG (IR-relevant in the Wilson sense).
- **Irrelevant directions**: couplings that flow to zero — these are the “cutoff artifacts”.
- **Marginal directions**: require more careful (perturbative) analysis.

2.3 Why the continuum limit is a second-order phase transition

Key connection: $\xi \rightarrow \infty$ is required for $a \rightarrow 0$ at fixed physical mass.

- In statistical mechanics, $\xi \rightarrow \infty$ signals a *second-order* (continuous) phase transition.
- First-order transitions: ξ remains finite — no continuum limit exists there.
- At the critical point the bare couplings $\{g_i^0\}$ must lie on the *critical surface* (the basin of attraction of the UV fixed point).

$$\xi(g_i) \sim |g_i - g_i^*|^{-\nu} \xrightarrow{g_i \rightarrow g_i^*} \infty. \quad (5)$$

2.4 Asymptotic freedom and the Gaussian fixed point (QCD context)

For non-Abelian gauge theories:

- The Gaussian (free-field) fixed point at $g = 0$ is UV-attractive: asymptotic freedom.

- The continuum limit is taken by tuning $g_0(a) \rightarrow 0$ as $a \rightarrow 0$, governed by the perturbative β -function:

$$a \frac{\partial g_0}{\partial a} = \beta(g_0) = -b_0 g_0^3 - b_1 g_0^5 - \dots \quad (6)$$

- Two-loop result gives the *Symanzik improvement* programme.

3 Scale Setting on the Lattice

3.1 The problem: lattice results are dimensionless

Monte Carlo generates numbers in units of a . To compare to experiment we must fix a in physical units.

- All dimensionful outputs come out as $O \cdot a^{d_O}$ for some dimension d_O .
- Need one experimental (or theoretical) input to fix a ; all other predictions are then genuine.

3.2 Common scale-setting schemes

3.2.1 Hadron spectrum

Use a well-known hadron mass (e.g. m_π , m_K , m_Ω , f_π) as input:

$$a = \frac{(m_\Omega)_{\text{lat}}}{m_\Omega^{\text{phys}}}. \quad (7)$$

3.2.2 The Sommer parameter r_0 / r_1

Defined from the static quark potential $V(r)$:

$$r^2 \frac{dV}{dr} \Big|_{r=r_0} = 1.65, \quad r_0 \approx 0.5 \text{ fm}. \quad (8)$$

Advantages: purely gluonic, computable at any β ; disadvantage: r_0 must itself be matched to experiment.

3.2.3 The gradient-flow scale t_0 / w_0

Modern approach using the Yang–Mills gradient flow:

$$\frac{\partial B_\mu}{\partial t} = D_\nu G_{\nu\mu}, \quad B_\mu|_{t=0} = A_\mu, \quad (9)$$

with scales defined by:

$$t^2 \langle E(t) \rangle|_{t=t_0} = 0.3, \quad t \frac{d}{dt} [t^2 \langle E \rangle] \Big|_{t=w_0^2} = 0.3. \quad (10)$$

- UV-finite at positive flow time; no renormalisation needed.
- Numerically precise and widely used in modern QCD simulations.

3.3 Discretisation errors and the continuum extrapolation

Results at finite a differ from the continuum by powers of a :

$$O(a) = O_{\text{cont}} + c_1 a^2 + c_2 a^4 + \dots \quad (\mathcal{O}(a) \text{ errors removed by Symanzik improvement}). \quad (11)$$

- Simulate at several values of a (several β), extrapolate to $a \rightarrow 0$.
- Improved actions (clover, staggered, twisted-mass, ...) remove leading a or a^2 terms.

3.4 Summary and outlook

- The lattice spacing a is removed by tuning bare couplings to a critical (UV fixed) point.
- This is precisely a second-order phase transition with diverging correlation length.
- Physical units enter through a single scale-setting observable; all further results are predictions.
- Next lecture: [*e.g. fermions on the lattice / Monte Carlo algorithms / hadron spectroscopy*].